SACUBITRIL + VALSARTAN

36 h gap from ACEi

Angioedema

CHEMICAL NAME Sacubitril: 4-{[(2S,4R)-1-(biphenyl-4-yl)-5-ethoxy-4-methyl-5-oxopentan-2-yl]amino}-4-oxobutanoic acid + valsartan (sodium salt complex)

CLASS Angiotensin receptor–neprilysin inhibitor (ARNI)

TRADE NAMES Entresto · Vymada · Uperio

THERAPEUTIC USES / INDICATIONS

- Chronic heart failure with reduced ejection fraction (NYHA II–IV); reduces cardiovascular death and hospitalisation (PARADIGM-HF).
- Selected HF with mildly reduced/preserved EF.

ADMINISTRATION

- **PO** twice daily with or without food.
- Stop ACE inhibitor at least 36 h before first dose (angioedema risk).
- Titrate every 2–4 weeks as tolerated.

DOSAGES

- **Start:** 49/51 mg twice daily (24/26 mg if ACEi-naïve, low BP, eGFR <30, moderate hepatic impairment, elderly).
- **Target:** 97/103 mg twice daily.
- Hold or reduce if SBP <95, K^+ >5.4.

PHARMACOKINETICS

- **Peak:** sacubitril 0.5 h, LBQ657 2 h, valsartan 1.5 h.
- Sacubitril is a prodrug converted to LBQ657 (active neprilysin inhibitor).
- Excreted in urine (LBQ657) and bile (valsartan). **t½:** 11 h (LBQ657), 10 h (valsartan).

PHARMACODYNAMICS

- **Neprilysin inhibition** → ↑ natriuretic peptides (BNP, ANP) → vasodilation, natriuresis, ↓ fibrosis.
- **AT1 blockade** (valsartan) counters RAAS activation.
- Note: BNP rises (neprilysin substrate) — use NT-proBNP to monitor.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hypotension (common), dizziness.
- Cough.
- Hyperkalaemia, renal impairment.

ADVERSE EFFECTS (SERIOUS)

- Angioedema (higher in Black patients, previous angioedema).
- Severe hypotension.
- Acute kidney injury.

CONTRAINDICATIONS & CAUTIONS

- Concurrent ACE inhibitor or within 36 h.
- History of angioedema (ACEi/ARB-related or hereditary).
- Pregnancy; severe hepatic impairment (Child-Pugh C).
- Aliskiren in diabetes.

KEY INTERACTIONS

- **ACE inhibitors:** contraindicated.
- **K^+-sparing diuretics, supplements:** hyperkalaemia.
- **NSAIDs:** AKI.
- **Statins:** ↑ statin levels (minor).

NURSING MANAGEMENT

- Confirm ACEi stopped ≥36 h earlier.
- Check BP, HR, U&E before each titration.
- Daily weights; signs of fluid overload or dehydration.
- Watch for angioedema.
- Use NT-proBNP (not BNP) for monitoring.

PATIENT EDUCATION

- Never take with an ACE inhibitor.
- Stand slowly; report dizziness, fainting.
- Report swelling of face, lips or tongue immediately.
- Weigh daily; report gain >2 kg in 3 days.

KEY POINTS

- ARNI — first-line HFrEF therapy.
- Hypotension, hyperkalaemia, angioedema.
- 36 h washout from ACEi.
- Monitor NT-proBNP, not BNP.

SOURCES CHECKED **[291]** Entresto 15 mg/16 mg granules in capsules for opening (UK SmPC) **[292]** Entresto, oral (US label) **[293]** sacubitril and valsartan, oral (US label)

Full references p. 432–457